RoleWalk: inductive structural roles for directed graphs from the characteristic function of random walks

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Abstract. Structural roles group nodes by their position in a graph rather than by proximity. RoleWalk follows a short random walk from each node and describes, at every step, two laws: that of the transition probabilities, through its characteristic function sampled at a few frequencies (how concentrated the walk is), and that of the walker's distance to its start (how far it has gone); optionally, low-order moments of a probabilistic curvature where it lands. The embedding is local, deterministic, reads directions and weights, and needs no training. We prove that it is invariant and local (so its coordinates mean the same in every graph), that its curvature-free distances are maximum mean discrepancies, that its characteristic-function block is Lipschitz in edge insertions, that it recovers triangle counts on regular graphs, so that it separates graphs 1-WL cannot (and conversely: the two are incomparable), and that distance-regular graphs with equal intersection arrays are a provable failure class. With a budget on the probability mass that pruning may drop, the approximation error is certified. Empirically it matches or beats struc2vec, GraphWave, Role2Vec, RESD and ReFeX on role classification at a fraction of the cost of most learned and diffusion baselines, has the best scores on a real directed-role dataset, transfers centralities to larger graphs as well as a dimension-matched top-k summary of the same walks and better than the other baselines, embeds a million nodes in 20 s to 2.5 min on one core (under 4 min with a certified 1% mass budget), and shows a preliminary improvement of GPS on ZINC when combined with RWSE.

1 Introduction

Claims, each backed by a result below:

1. All of: inductive, directed, weighted, training-free, deterministic, online, fast (Table 1).
2. Accuracy competitive with the state of the art on role classification (clearly ahead on one of six networks) at a fraction of the cost; a million nodes in minutes on one core (Table 2, Figure 1, Table 13).
3. Reads edge directions natively, including walks that change direction; highest scores on both wiki-RfA tasks, best retrieval and competitive clustering on synthetic directed motifs; on food webs, not distinguishable from local statistics or Digraphwave (Table 5, Table 6, Table 7, Proposition 10).
4. Pruning with a certified error (Proposition 13).
5. Transfer: a regressor fitted on one graph predicts path-based centralities of another from the same generator four times larger as well as the top-k probabilities of the same walks (a tie: the law of the distance carries most of it) and better than every other baseline; under a change of generator parameters it stays ahead of every embedding, but local statistics transfer better on average (Table 10, Section 6.4; Proposition 1).
6. Separates graphs that 1-WL cannot with 3 walk steps (the two are incomparable), close to 16-step RWSE on BREC; a provable failure class (Section 4, Table 8).
7. Useful as a structural encoding for graph transformers (Table 12).

2 Related work

Feature-based roles: ReFeX [1], RolX [2]. Walks and word2vec: struc2vec [3], Role2Vec [4]; directed and weighted extension of struc2vec, struc2vec++ [5] (not evaluated here). Diffusion: GraphWave [6], closest to RoleWalk (Proposition 11), and Digraphwave [7], its extension to directed and weighted graphs (heat diffusion on the out- and in-graph, timescales read from each graph, characteristic functions aggregated over neighbours). Deep role embeddings: RESD [8], from the RONE survey and benchmark [9], whose code and Wikipedia talk datasets we use. Transferable structural signatures: xNetMF in REGAL [10], from degree

distributions of neighbourhoods, with landmark-based coordinates fitted on the graphs to align. Structural encodings for GNNs: RWSE [11], used by GPS [12]. Expressiveness: 1-WL [13], [14], [15], CSL [16], BREC [17]. Curvature: Ollivier-Ricci [18], [19]. Kernel mean embeddings and MMD: [20].

3 Method

Walk. $G = (V, E)$ with non-negative weights A (directed or not, self-loops ignored), s_u the weighted out-degree and d_u the number of out-neighbours with positive weight, $b_u = s_u/d_u$ ($b_u = 1$ if $d_u = 0$), $B = \mathrm{diag}(b)$, and the self-loop-augmented walk

$$H = \mathrm{diag}\left((A+B)\mathbf{1}\right)^{-1}(A+B), \quad H_{uu} = \frac{1}{d_u+1}, \quad H_{uv} = \frac{d_u}{d_u+1}\frac{A_{uv}}{s_u} \quad (d_u > 0). \tag{1}$$

On unweighted graphs $B = I$ and $H = \tilde{D}^{-1}(I + A)$ with $\tilde{D} = \mathrm{diag}((I + A)\mathbf{1})$; on weighted ones the self-loop weighs an average edge, so H does not depend on the unit of the weights. For a node u and a step w, $p_u^w = e_u^\top H^w$ is the distribution of the walk after w steps. Directed graphs use the out-walk H_+ (A) and the in-walk H_- ($A^\top$), concatenate both embeddings, and add the characteristic functions of the rows of H_+H_- (one step forward, one back: nodes sharing targets with u) and H_-H_+ (shared sources), the *mixed probes* (Proposition 10): 178 features, 190 with curvature. On undirected graphs $H_+ = H_-$ and the probes would repeat H^2, so they are left out.

Characteristic function. With $f_\theta(x) = e^{i\theta x} - 1$ and a grid $\Theta = \{\theta_1, ..., \theta_m\}$ (log-spaced in $[1, 100]$, $m = 10$),

$$\varphi_u^w(\theta) = \sum_{j \in V} f_\theta\left(p_{uj}^w\right), \quad \theta \in \Theta. \tag{2}$$

Since $f_\theta(0) = 0$, the sum runs over the support of p_u^w only.

Distance of the walker. With $d(u, j)$ the length of the shortest walk from u to j and X_w the walker's position after w steps,

$$s_u^w(k) = P(d(u, X_w) = k) = \sum_{j : d(u,j) = k} p_{uj}^w, \quad k = 0, ..., w, \tag{3}$$

is the law of the walker's distance to its start, which the multiset of probabilities ignores: whether the walk stays near u (loops, triangles) or moves out. It takes $w + 1$ values, so these $w + 1$ numbers describe it exactly. We call the sets of nodes at each distance *onion layers*. Supports of earlier powers give the distances, so no extra walk is computed.

At each step, RoleWalk thus describes two laws of the same walk: how concentrated it is (the characteristic function of its probabilities) and how far it has gone (the law of its distance). Both are integrals over one measure, under one kernel (Proposition 3).

Curvature (option). For an edge xy of the symmetrised unweighted graph, $\kappa(x, y) = \sum_z \min\left(m_x(z), m_y(z)\right)$ with m_x the one-step lazy walk from x; κ_x is its mean over the edges at x. Each step adds the mean $\mu_u^w = \sum_j p_{uj}^w \kappa_j$ and the standard deviation σ_u^w of κ under p_u^w. Since a unit of mass that does not overlap moves at least one step, $W_1\left(m_x, m_y\right) \geq 1 - \kappa(x, y)$: κ is an upper bound of Ollivier-Ricci curvature $1 - W_1$, made of probabilities.

One measure, one kernel. Every node j reached at step w is an atom marked by its step and its layer, which gives the *footprint* of the walks from u:

$$\mu_u = \sum_{w=1}^{L} \sum_{j : p_{uj}^w > 0} \delta_{\left(w, p_{uj}^w, d(u,j)\right)}. \tag{4}$$

The embedding is the integral of a single feature map over the footprint (a finite measure, not normalised), $X_u = \int \Psi \, d\mu_u$, with

$$\Psi(w, a, \ell) = e_w \otimes (\psi(a) \oplus \|\Theta\| a e_\ell), \quad \psi(a) = \left(f_\theta(a)\right)_{\theta \in \Theta}, \tag{5}$$

which gives, for each step $w = 1, ..., L$ ($L = 3$), $\operatorname{Re}\varphi_u^w$, $\operatorname{Im}\varphi_u^w$ and $\|\Theta\| s_u^w$. With curvature the mark is $\big(d(u,j), \kappa_j\big)$ and Ψ adds $\|\Theta\| a(\kappa, \kappa^2)$; we report $\|\Theta\|(\mu_u^w, \sigma_u^w)$, which carries the same information but, being nonlinear, is not an integral of Ψ: kernel statements below are for the curvature-free embedding (or raw moments). In all, $2mL + \sum_{w \le L}(w+1)(+2L) = 60 + 9(+6) = 69(75)$ features: the characteristic function, then the law of the distance at each step. The probability block ignores the distance; the distance block is the first-order part of one characteristic function per layer, which is what fixes the factor $\|\Theta\| = \big(\sum_\theta \theta^2\big)^{1/2}$ (Proposition 3). On directed graphs the step mark w becomes a block mark b, a direction and a step $(\pm, w)$ or a mixed probe $(+-, -+)$, and mixed-probe atoms carry no layer: $\Psi(b, a) = e_b \otimes \psi(a)$. Equivalently, with the centred joint Fourier transform of the footprint $F_u^w(\theta, \omega) = \sum_j e^{i\omega d(u,j)}\big(e^{i\theta p_{uj}^w} - 1\big)$ (unnormalised: $F(0,0) = 0$; not the law of $\big(p_{uX_w}, d(u, X_w)\big)$, which would weight atom j by p_{uj}), the characteristic function is $F_u^w(\theta, 0)$ and the law of the distance is its derivative at $\theta = 0$, $-i\partial_\theta F_u^w(0,\omega) = \sum_k s_u^w(k) e^{i\omega k}$; with $N = w + 1$, the values $\tilde{g}_j = \big(-i/\sqrt{N}\big)\partial_\theta F_u^w(0, 2\pi j/N)$ form the unitary DFT of s_u^w, so $\|\tilde{g}\|_2 = \|s_u^w\|_2$: they determine s_u^w with the same Euclidean geometry. Without it the layers carry 0.02% of the variance and do not change the roles found by k-means (Section 6.6).

Pruning. After each product, entries below $\mathrm{tol} = 10^{-4}$ are dropped; their mass is added back to $\operatorname{Im}\varphi$ to first order ($\theta \cdot$ mass), the same tangent approximation as the layer block. The first step is never pruned. Dropped mass can gather again: in a star with more than 10^4 leaves, the leaf mass of step 2 falls below tol although half of it returns to the centre at step 3. With a budget δ (`max_missing`), a row drops its small entries only while its missing mass $r_u^w = 1 - \|q_u^w\|_1$ stays below δ, which bounds the error (Proposition 13). Distances are read from supports before pruning, as the step at which each node is first reached: its distance when $\mathrm{tol} = 0$; with pruning, for $L \le 3$ every retained entry still gets its exact distance, while layer masses lose what pruning drops; beyond three steps, a node first reached through a pruned entry is labelled later than its distance. Both implementations use this convention. A row whose small entries would exceed the budget keeps all of them. Results in Section 4 are otherwise stated for $\mathrm{tol} = 0$.

Implementation. For each root, one pass propagates a sparse row with a dense accumulator, prunes it, and summarises it (characteristic function, layers from the step at which each node is first reached, curvature moments) without storing the powers: $O(n)$ memory per thread, and rows are independent. Mixed probes are sparse products computed by blocks of rows whose work (entries before pruning) is bounded by a budget plus at most one row's own work. Below $p = 10^{-3}$ every angle θp is under 0.1, and the characteristic function comes from the power sums $\sum p^k$, $k \le 8$ instead of $2m$ sines (truncation error below $\theta^9 p_{\max}^8 / 9! < 3 \cdot 10^{-12}$ per coordinate and row in exact arithmetic; outputs are float32).

Roles. Ward clustering with silhouette for the number of roles, or k-means with a given number; fitted centroids label nodes of any graph.

4 Properties

Throughout, L is the number of steps, $B_r(u)$ the subgraph induced by nodes within distance r of u (out- and in-distance for directed graphs; with curvature, or on directed graphs, distance in the underlying undirected graph, which contains everything these read), graphs have finite non-negative weights and no self-loops, distances are taken in the graph of positive-weight edges, m_x and κ are defined on the simple unweighted symmetrisation ($\kappa = 0$ at isolated nodes), and statements hold for $\mathrm{tol} = 0$, in exact arithmetic.

Proposition 1 (locality and invariance)**.** *X_u is determined by the isomorphism class of the rooted graph $(B_r(u), u)$, with $r = L$ for pure walks, $r = \max(L, 2)$ (undirected distance) on directed graphs with mixed probes, and $r = L + 2$ with curvature. In particular it does not depend on $|V|$, nodes related by an automorphism get equal embeddings, and a node can be embedded from its neighbourhood alone.*

Proof. $p_{uj}^w = \sum_{u = x_0, \dots, x_w = j} \prod_t H_{x_t x_{t+1}}$ runs over walks of length $w \le L$ from u. Their nodes lie within distance w, and H_{xy} needs the edges at x (and s_x, d_x), with x within distance $w - 1$: all are in $B_L(u)$. An isomorphism of rooted balls maps these walks bijectively and preserves their weights, so it maps p_u^w to $p_{u'}'^w$ entry by entry; Equation 2 is a sum over j, hence equal. Distances up to L are read inside $B_L(u)$. With curvature, κ_j for j within distance L needs the closed neighbourhoods of j and of its neighbours, inside $B_{L+2}(u)$. A mixed probe goes from u to a neighbour x (one direction) then to a neighbour of x (the other): it needs the edges at u and at x, all within undirected distance 2 of u, including the normalisation of the second step at x. □

Coordinates therefore mean the same thing in every graph, which makes transfer (Table 10, Table 11) well defined; whether it succeeds is an empirical question.

Proposition 2 (injectivity)**.** *Let P, Q be finite multisets of reals in $(0, 1]$. If $\sum_{p\in P} f_\theta(p) = \sum_{q\in Q} f_\theta(q)$ for all θ in a set with an accumulation point (e.g. $[1, 100]$), then $P = Q$. For a finite grid, given any countable family of pairs $P \neq Q$, Lebesgue-almost every $\Theta \in \mathbb{R}^m$ separates all of them.*

Proof. $F(\theta) = \sum_P e^{i\theta p} - \sum_Q e^{i\theta q} - (|P| - |Q|)$ is entire and vanishes on a set with an accumulation point, hence everywhere. Exponentials $e^{i\theta\lambda}$ with distinct λ are linearly independent, so each $\lambda > 0$ has the same multiplicity in P and Q (and $|P| = |Q|$, coefficient of $\lambda = 0$). For the second claim, the real zeros of each nonzero entire function F are countable; a countable union of countable sets is null, so almost every θ_1 already separates every pair. □

So with Θ an interval, φ_u^w is exactly the multiset of nonzero transition probabilities of u after w steps. The second claim is qualitative: the separation margin, not its existence, motivates $m = 10$ frequencies across two decades.

Proposition 3 (one kernel on the footprint)**.** *Without curvature, $\|X_u - X_v\|^2 = \mathrm{MMD}_K^2(\mu_u, \mu_v)$ (extended to finite measures) for the positive-semidefinite kernel on marked atoms*

$$K((w, a, \ell), (w', b, \ell')) = \delta_{ww'}\big[k_\Theta(a, b) + \|\Theta\|^2 ab\,\delta_{\ell\ell'}\big], \tag{6}$$

and on directed graphs, with block marks, for $K((\beta, a, \ell), (\beta', b, \ell')) = \delta_{\beta\beta'}\big[k_\Theta(a, b) + \mathbb{1}_{\beta\text{ pure}}\|\Theta\|^2 ab\,\delta_{\ell\ell'}\big]$,

$$k_\Theta(a, b) = \sum_{\theta\in\Theta}[\cos\theta(a - b) - \cos\theta a - \cos\theta b + 1]. \tag{7}$$

The first term compares probabilities whatever their layer, the second compares masses in the same layer. Moreover, with $t = \max(\Theta)\max(a, b)$,

$$\big|k_\Theta(a, b) - \|\Theta\|^2 ab\big| \le \|\Theta\|^2 ab\, t^2(7/12 + t^2/36), \tag{8}$$

so the layer term is the first-order part of $k_\Theta(a, b)\,\delta_{\ell\ell'}$, the kernel of one characteristic function per layer, and $\|\Theta\|$ is the only positive factor for which the two agree to first order (with orthonormal layer coordinates). The distance is an MMD pseudometric: a finite grid is not proved characteristic.

Proof. $X_u = \sum \Psi$ over the atoms of μ_u, so X_u is the kernel mean embedding of μ_u and $\|X_u - X_v\|^2$ is the MMD [20] for $K = \mathrm{Re}\langle\Psi, \Psi'\rangle$, positive semidefinite as an inner product. The blocks e_w and e_ℓ are orthonormal, which gives $\delta_{ww'}$ and $\delta_{\ell\ell'}$; expanding $\mathrm{Re}\big(e^{i\theta a} - 1\big)\overline{(e^{i\theta b} - 1)}$ gives k_Θ. Unreached nodes have $\Psi = 0$ and do not count. For the bound, fix θ and write the real part as $(\cos\theta a - 1)(\cos\theta b - 1) + \sin\theta a \sin\theta b$. Since $|\cos x - 1| \le x^2/2$, the first product is at most $\theta^4 a^2 b^2/4 \le \theta^2 ab\, t^2/4$. With $\sin\theta a = \theta a + r_a$ and $|r_a| \le \theta^3 a^3/6$, $\big|\sin\theta a \sin\theta b - \theta^2 ab\big| \le \theta^4 ab(a^2 + b^2)/6 + \theta^6 a^3 b^3/36 \le \theta^2 ab\, t^2(1/3 + t^2/36)$. Summing over θ gives the bound, and $\|\Theta\|^2 = \sum_\theta \theta^2$ is the coefficient of ab. □

The concatenation is therefore a sum of kernels on one object: a mark-blind term on probabilities and a mark-aware term on masses. The first-order regime $t \ll 1$ holds where walks spread thin, not for low-degree nodes nor for returns to a hub ($\max(\Theta) = 100$): the linear layer kernel is a choice, $L(L + 3)/2 = 9$ coordinates instead of $mL(L + 3) = 180$ for one characteristic function per layer, and the full per-layer kernel is not consistently better despite its larger dimension: it helps on synthetic and directed motifs and hurts airport transfer (Table 15). Further marks enter the same way, as factors of the second term: curvature, or labels on the steps of each path.

Proposition 4 (stability)**.** *Let G be simple, undirected and unweighted, and G' be G plus an edge ab, with $d = \min(d_a, d_b)$ the smaller endpoint degree in G. For every node u and step w,*

$$\|\varphi_u^w(G') - \varphi_u^w(G)\|_2 \le w\|\Theta\|_2 \cdot \frac{2}{d + 2}, \tag{9}$$

and the left side is 0 if u is at distance $\ge w$ from both a and b.

Proof. $|e^{i\theta x} - e^{i\theta y}| \le \theta\,|x - y|$, so $\|\varphi_u^w(G') - \varphi_u^w(G)\|_2 \le \|\Theta\|_2 \|p_u'^w - p_u^w\|_1$. Telescoping, $H'^w - H^w = \sum_{t=0}^{w-1} H'^t(H' - H)H^{w-1-t}$; stochastic matrices have $\|\cdot\|_\infty = 1$, so each row of the difference has ℓ_1 norm at most $w\|H' - H\|_\infty$. Only rows a and b of H change; row a goes from $d_a + 1$ entries $1/(d_a + 1)$ to $d_a + 2$

entries $1/(d_a+2)$, an ℓ_1 change of $2/(d_a+2)$. Row u of $H'^t(H'-H)$ is nonzero only if a walk of $t \le w-1$ steps from u reaches a or b. □

Insertions at high-degree nodes barely move the characteristic-function block, and the effect is confined to the $(L-1)$-ball. On 200 random insertions (powerlaw-cluster graphs, 60 nodes) the largest observed move of this block is 0.52 of the bound (`theory_checks.py`). The bound does not extend to the layer block, which reads distances: in $K_{2,d}$, joining the two hubs a, b moves $s_a^2(2)$ from $d/(3(d+1))$ to 0, whatever d.

Proposition 5 (degree and triangles)**.** *Let G be simple, undirected and unweighted. Then $s_u^1(0) = 1/(d_u+1)$, and if G is d-regular ($d>0$), $s_u^3(0) = (H^3)_{uu} = (1+3d+2t_u)/(d+1)^3$, with t_u the number of triangles at u; already at step 2, $s_u^2(1) = (2d+2t_u)/(d+1)^2$. Hence RoleWalk with $L \ge 2$ separates 1-WL-equivalent graphs, e.g. two triangles and a hexagon, or any two d-regular graphs with different triangle counts.*

Proof. $(I+A)^3 = I + 3A + 3A^2 + A^3$ with $(A)_{uu} = 0$, $(A^2)_{uu} = d$, $(A^3)_{uu} = 2t_u$, and $H = (I+A)/(d+1)$. At step 2, the mass on the neighbours of u is $\sum_{j\sim u} (I+2A+A^2)_{uj}/(d+1)^2$, with $\sum_{j\sim u} A_{uj} = d$ and $\sum_{j\sim u} (A^2)_{uj} = 2t_u$. 1-WL gives every node of two d-regular graphs of the same size the same colour. □

Proposition 6 (relation to RWSE)**.** *The distance-0 layer $s_u^w(0) = (H^w)_{uu}$ is the return probability of the lazy walk, i.e. RWSE [11] for H, up to L steps. On a simple, undirected, unweighted d-regular graph with $d>0$, $\left\{(H^w)_{uu}\right\}_{w\le L}$ and the simple-walk RWSE $\left\{(P^k)_{uu}\right\}_{k\le L}$, $P = A/d$, determine each other.*

Proof. $H = (I+dP)/(d+1)$, so $(H^w)_{uu} = (d+1)^{-w}\sum_{k=0}^{w}\binom{w}{k} d^k (P^k)_{uu}$: a triangular system with nonzero diagonal $d^w/(d+1)^w$. □

On irregular graphs the binomial transformation above does not apply; RoleWalk adds the other layers and the full distribution of transition probabilities.

Proposition 7 (a failure class: distance-regular graphs)**.** *Let G, G' be unweighted distance-regular graphs with the same intersection array. Then all nodes of G and G' get the same embedding, for every L, with or without curvature. This covers strongly regular graphs with equal parameters (diameter 2).*

Proof. Let A_i be the distance-i matrices. Distance-regularity gives $AA_i = b_{i-1}A_{i-1} + a_iA_i + c_{i+1}A_{i+1}$ with coefficients from the intersection array [21], so by induction $(I+A)^w = \sum_i \alpha_{w,i}A_i$ with $\alpha_{w,i}$ depending only on the array. As $H = (I+A)/(k+1)$, $p^w_{uj} = \alpha_{w,d(u,j)}/(k+1)^w$: each row has value $\alpha_{w,i}/(k+1)^w$ with multiplicity k_i, the number of nodes at distance i, fixed by the array. So φ_u^w and $s_u^w(i) = k_i\alpha_{w,i}/(k+1)^w$ are the same for all nodes of both graphs. For an edge xy, $\kappa(x,y) = |N[x] \cap N[y]|\ /(k+1) = (a_1+2)/(k+1)$, a constant, so μ_u^w is constant and $\sigma_u^w = 0$. □

Checked on Shrikhande / 4×4 rook, Chang / $T(8)$ and the diameter-3 pair Doob $D(1,1)$ / Hamming $H(3,4)$: largest difference between any two nodes $\le 3\cdot 10^{-7}$ (float32). This explains the 0/20 distance-regular pairs of BREC (Table 9); RWSE, which is constant on these graphs for the same reason, fails them too.

Proposition 8 (locality limits graph-level separation)**.** *Let A and B be a 40-cycle with pendant nodes attached at cycle positions $\{0,15\}$ and $\{0,20\}$. They are not isomorphic and 1-WL separates them, but RoleWalk with $L=3$ (with curvature) does not.*

Proof. The pendants are more than $2(L+2)$ apart in both graphs, so every rooted ball $B_{L+2}(u)$ contains at most one pendant and is fixed by u's distance to it; both graphs have the same multiset of balls. Conclude by Proposition 1. For 1-WL, let h mark the two degree-3 nodes, which round 1 identifies; round $t+1$ colours determine $(A^t h)_v$, the number of t-step walks from v to a marked node. Both graphs are bipartite (an even cycle with pendant trees). In A, the only 15-step walk from node 0 to node 15 is the short arc, and no odd walk returns to 0, so $(A^{15}h)_0 = 1$; in B, nodes 0 and 20 are on the same side, so $(A^{15}h)_0 = 0$. After 16 rounds the marked nodes have different colours. □

Proposition 9 (RoleWalk and 1-WL are incomparable)**.** *Proposition 5 gives graphs that RoleWalk separates and 1-WL does not. Conversely, without curvature, for $L \in \{2,3\}$ there are nodes that L-round 1-WL (uniform initial colours) separates and RoleWalk with L steps does not, for every set of frequencies.*

Proof. Take a clique K_5 on $\{r,a,b,c,d\}$ and ten further nodes, adjacent to b,c,d in G, and five to a,b and five to c,d in G'. The neighbour degrees of r are $\{4,14,14,14\}$ in G and $\{9,9,9,9\}$ in G', so two rounds separate r. Walks from r agree: at step 1 each clique node has probability $1/5$; at step 2 each clique node has $1/5(1/5+2/5) = 3/25$ (the inverse augmented degrees of the neighbours of r sum to $1/5+3/15 = 2/5$ in G

and $4/10 = 2/5$ in G') and each outer node $1/25 = 1/5 \cdot 3/15 = 1/5 \cdot 2/10$, with the same layers. For $L = 3$, add a leaf z at r and root at z: three rounds separate z. From z, step 1 puts $1/2$ on z and r; step 2 puts $1/3$ on z and r and $1/12$ on each of a, b, c, d; step 3 puts $2/9$ on z, $23/90$ on r, $4/45$ on each of a, b, c, d and $1/60$ on each outer node, in both graphs, with layers $(2/9, 23/90, 16/45, 1/6)$ (exact rational computation, `theory_checks.py`). With curvature, both pairs are separated. These are node-level collisions; Proposition 8 gives a graph-level one. □

On the 6,781 nodes of the 996 connected graphs with at most 7 nodes, no pair separated by 3-round 1-WL is confused by RoleWalk ($L = 3$), which splits 571 of the 1-WL classes (embeddings grouped by single linkage at a sup-distance below 10^{-4} of the largest coordinate, i.e. connected components of that graph; `theory_checks.py`); the counterexamples above have 15 and 16 nodes. RoleWalk is also local, which 1-WL with unbounded rounds is not.

Proposition 10 (walks that change direction)**.** *Let u be a source pointing to two sinks (G), or one of the two sources of a directed $K_{2,2}$ (G'). For every L, with or without curvature, u gets the same out- and in-walk embedding in G and G'; but $e_u^\top H_+ H_-$ has values $(2/3, 1/6, 1/6)$ in G and $(5/9, 2/9, 1/9, 1/9)$ in G'; at $\theta = 1$ (in the default grid) $\operatorname{Re}\varphi$ is -0.2418 and -0.1873, so the mixed probes separate them.*

Proof. Out-walk: u keeps $1/3$ and sends $1/3$ to each sink, where it stays; the layers agree. In-walk: u has no in-edge and stays. The symmetrised graphs are a path $a - u - b$ and a 4-cycle; no edge has a common neighbour, and every edge has $\max(d_x, d_y) = 2$, so every node has $\kappa = 2/3$ in both. In H_-, a sink with in-neighbours N spreads uniformly over itself and N: $1/2$ on itself and u in G, $1/3$ on itself, u and the other source in G'. Hence $e_u^\top H_+ H_- = 1/3\big(e_u + \sum_{\text{sinks } s} e_s^\top H_-\big)$, which gives the two rows. □

The out- and in-walks see where u sends and receives, not who shares its targets (voters of the same candidates) or its sources; Table 5 and Table 6 measure what this adds.

Proposition 11 (relation to GraphWave)**.** *GraphWave [6] embeds u by $\chi_u(t) = \frac{1}{n}\sum_j e^{it\Psi_{ju}}$ with $\Psi = U g_s(\Lambda) U^\top$ the heat kernel at a scale s chosen from the spectrum of each graph. Then $n(\chi_u(t) - 1) = \sum_j f_t(\Psi_{ju})$: Equation 2 with the heat kernel in place of H^w. χ_u depends on n and on s, so the same rooted neighbourhood can get different coordinates in graphs of different sizes or spectra.*

Proof. $\sum_j \big(e^{it\Psi_{ju}} - 1\big) = n\chi_u(t) - n$. Ψ has full support on a connected graph, so Proposition 1 fails; $1/n$ and s are graph-level. □

This explains most of why GraphWave transfers at chance level: without its $1/n$, it reaches 0.455 on airport transfer, against 0.488 for RoleWalk (Table 11).

Proposition 12 (cost)**.** *Let q_u^t be the retained (pruned) row after step t. Step w costs $O\Big(\sum_u \sum_{j \in \text{ supp } q_u^{w-1}} (d_j + 1)\Big)$; with threshold pruning (*$\text{tol} > 0$*, no budget), $|\text{supp } q_u^t| \le \min(|B_t(u)|, \lfloor 1/\text{ tol} \rfloor)$ for $t \ge 2$, while the first step keeps all $d_u + 1$ entries and a budget can keep any number. The cap is on retained rows, not on the accumulator before pruning. The characteristic function adds $O(m \cdot \text{nnz})$, the layers $O(\text{nnz})$. As $\kappa(x, y) = (c_{xy} + 2)/(\max(d_x, d_y) + 1)$ with c_{xy} the common neighbours of x and y, curvature costs a triangle enumeration, $O(|E|^{3/2})$ plus $O(n \log n + |E|)$ set-up (on a subset of nodes, $O(\min(d_x, d_y) \log d)$ per edge at those nodes); this is the compiled implementation, whose fallback without a compiler uses one sparse product, quadratic on a star. The embedding of a set S depends only on its ball (Proposition 1), and only its walks are computed; the transition matrices are still built for the whole graph.*

Proof. Entries are non-negative and each row sums to at most 1, so at most $1/\text{ tol}$ of them are $\ge \text{tol}$. Both closed neighbourhoods are uniform ($1/(d+1)$ on $d+1$ nodes) and share x, y and their common neighbours, so the overlap is $c_{xy} + 2$ times the smaller mass; orienting edges by degree enumerates each triangle once, scanning at most $\sqrt{2\,|E|}$ forward neighbours per node: $O(|E|^{3/2})$. The rest follows from sparse products and Proposition 1. □

Proposition 13 (certified pruning)**.** *For one walk direction, without curvature or mixed probes, $L \le 3$ and in exact arithmetic, let $\hat{X}_u$ be computed with pruning and r_u^w the mass row u misses at step w. Then*

$$\left\| X_u - \hat{X}_u \right\|_2 \le \sqrt{5} \|\Theta\|_2 \left(\sum_{w \le L} \left(r_u^w\right)^2 \right)^{1/2}, \tag{10}$$

so with a budget δ, $\left\|X_u - \hat{X}_u\right\|_2 \le \sqrt{5L}\|\Theta\|_2\delta$ ($\sqrt{10L}\|\Theta\|_2\delta$ for the two directions of a directed graph). Mixed probes with missing masses r_β add $4\|\Theta\|_2^2 \sum_\beta r_\beta^2$ to the squared bound.

Proof. Propagation and deletion keep entries non-negative, so the pruned row q_u^w satisfies $0 \le q_u^w \le p_u^w$ entrywise and $\|p_u^w - q_u^w\|_1 = r_u^w$. For each θ, $\left|\sum_j f_\theta(p_j) - f_\theta(q_j)\right| \le \theta r_u^w$, and the first-order correction $i\theta r_u^w$ adds at most θr_u^w: the characteristic-function block moves by at most $2\|\Theta\|_2 r_u^w$. Layers use the exact distances (supports read before pruning are exact up to step 2, and every node reached at step 3 outside them is at distance 3), so each layer loses $\sum_{j\in \text{ layer}}(p_j - q_j) \ge 0$, in total r_u^w: the scaled layer block moves by at most $\|\Theta\|_2 r_u^w$. Summing squares over blocks and steps gives the bound; a budget enforces $r_u^w \le \delta$, since propagation by a stochastic matrix keeps the missing mass and a row only drops entries while it stays below δ. Directions are orthogonal blocks; a mixed probe is a characteristic function without layers, hence $2\|\Theta\|_2 r_\beta$. □

The bound is a posteriori without a budget ($r_u^w \le 1$) and a priori with one; it is absolute in feature space ($\|\Theta\|_2 \approx 125$, so about 24 at $\delta = 0.05$, $L = 3$), not a guarantee on downstream accuracy. The certificate is an exact-arithmetic pruning certificate, returned per node by the library (`certificate_`, with the missing masses `missing_`); it is not a numerical certificate of the returned array, which also carries the Taylor truncation of the fused kernel and float32 rounding, $\left\|X_u - \hat{X}_u^{\text{returned}}\right\| \le B_{\text{prune}} + \varepsilon_{\text{Taylor}} + \varepsilon_{\text{float}}$, and curvature moments are not covered.

5 Experiments

All CPU timings on one core. Classification: logistic regression on standardised embeddings, 50 stratified 80/20 splits. Best in bold, second underlined, among embeddings; the degree baseline (log-degree alone, which shows how far labels follow activity) is set apart and not ranked. Code and CSV files in `benchmarks/`.

Table 1: Properties, checked against each paper: ✓ yes, ✗ no, ~ partial.

	Local, fixed coords	Directed	Weighted	No training	Online	Deterministic	Time (USA)
RoleWalk	✓	✓	✓	✓	✓	✓	0.2 s
struc2vec	✗	✗[a]	✗[a]	✗	✗	✗	564 s
GraphWave	✗[c]	~	✓	✓	✗	✓	18 s
Role2Vec	~	✓[b]	✗	✗	~	✗	10 s
RESD	✗	✗	✗	✗	✗	✗	8 s
ReFeX	~	~	~	✓	~	✓	0.1 s
Digraphwave	~	✓	✓	✓	~	✓	1.1 s
RWSE	✓	~	✓	✓	✓	✓	1.7 s

[a] struc2vec++ [5] extends it. [b] In the paper; the reference implementation is undirected. [c] Scale chosen from the spectrum and $1/n$ factor (Proposition 11). "No training": struc2vec and Role2Vec fit skip-gram models, RESD a variational autoencoder.

Table 2: Role classification accuracy. struc2vec is unseeded: mean ± std over 5 runs. The random-walk heat-kernel CF is RoleWalk's characteristic function on the rows of $\exp(s(P-I))$, $s \in \{1,2,3\}$, with the same frequencies: fixed scales and no $1/n$, but a random-walk rather than GraphWave's spectral heat kernel. Top-k + layers: the 20 largest probabilities at each step instead of the characteristic function, same layers, same dimension. Local statistics: see Table 10. `roles.csv`, `roles_gw.csv`, `roles_matched.csv`, `struc2vec_runs.csv`.

	Brazil	Europe	USA	Reality-call	Actor	Film*
RoleWalk	0.743	0.583	<u>0.657</u>	<u>0.860</u>	<u>0.487</u>	0.569
RoleWalk + curv.	0.741	<u>0.596</u>	**0.666**	<u>0.860</u>	0.485	0.581
struc2vec	0.705 ± .022	0.561 ± .020	0.596 ± .004	–	–	–
RESD	0.728	0.544	0.653	0.857	0.476	0.563
GraphWave	**0.769**	0.561	0.587	<u>0.860</u>	0.481	–
GraphWave, no $1/n$	**0.769**	0.560	0.587	<u>0.860</u>	0.481	–
Random-walk heat-kernel CF	<u>0.750</u>	**0.598**	0.633	0.859	0.484	0.549
Digraphwave	0.724	0.593	0.641	0.859	**0.488**	<u>0.606</u>
Top-k + layers	0.672	0.590	0.625	0.858	0.482	0.564
Local statistics	0.716	0.584	0.616	**0.862**	0.486	0.517
ReFeX	0.691	0.576	0.622	0.851	0.478	0.579
Role2Vec	0.476	0.366	0.562	0.844	0.399	**0.743**
RWSE	0.711	0.572	0.517	0.855	0.363	0.450
Degree (baseline)	*0.756*	*0.554*	*0.567*	*0.858*	*0.476*	*0.478*

* A node-type task (Section 6.7). "–": too slow or out of memory. Paired differences: Table 3. Within one graph, GraphWave without $1/n$ is GraphWave up to an affine map that standardisation cancels: the two rows differ by numerical noise. Appending the local statistics to RoleWalk gives 0.752 / 0.595 / 0.659 / 0.861 / 0.486 / 0.598.

Table 3: RoleWalk + curv. minus each method, accuracy, paired over the 50 splits of Table 2; bold: significant at 5% with the Nadeau–Bengio correction for overlapping splits, without correction for the 10 comparisons per dataset. The splits are of one graph, not independent graphs. `paired_tests.csv`.

	Brazil	Europe	USA	Reality-call	Actor	Film
RoleWalk	−0.002	+0.012	+0.008	−0.000	−0.002	**+0.012**
GraphWave	−0.028	+0.034	**+0.079**	+0.000	+0.004	–
RW heat CF	−0.009	−0.003	**+0.032**	+0.001	+0.001	**+0.032**
RESD	+0.013	+0.051	+0.013	+0.003	+0.010	**+0.018**
ReFeX	+0.050	+0.019	**+0.044**	**+0.008**	+0.007	+0.003
Top-k + layers	**+0.069**	+0.005	**+0.041**	+0.002	+0.003	**+0.017**
Local statistics	+0.024	+0.011	**+0.050**	−0.002	−0.001	**+0.065**
Role2Vec	**+0.265**	**+0.229**	**+0.104**	**+0.016**	**+0.087**	**−0.162**
RWSE	+0.030	+0.023	**+0.148**	**+0.004**	**+0.122**	**+0.131**
Degree	−0.016	+0.041	**+0.099**	+0.001	+0.009	**+0.103**

Intervals are about ±0.05 on Brazil and Europe (131 and 399 nodes).

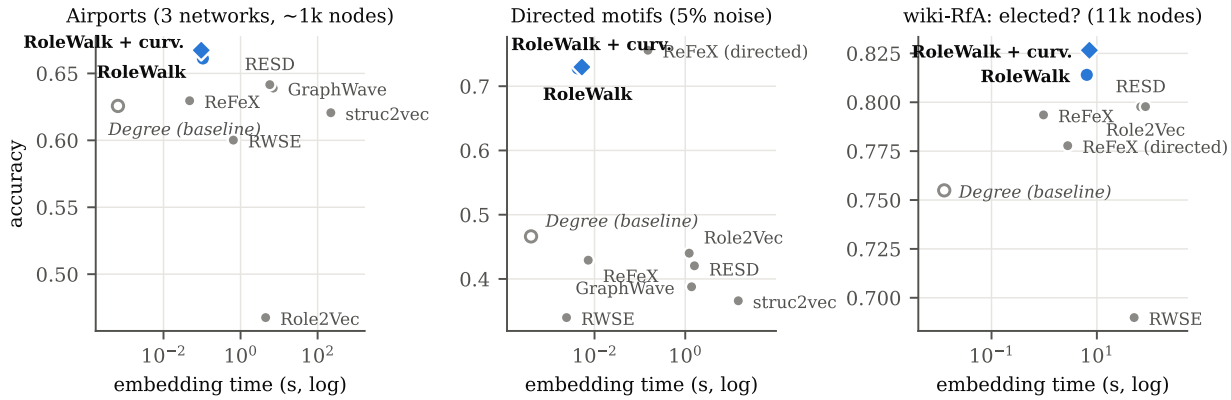


Figure 1: Accuracy against embedding time (one core, log scale), every method that runs on each benchmark. Airports: mean over Brazil, Europe and USA (struc2vec: mean of 5 runs). Directed motifs: 5% rewired edges, mean of 3 seeds. wiki-RfA: task A of Table 6. Hollow marker: degree, a trivial baseline rather than an embedding. On directed graphs RoleWalk includes the mixed probes. RoleWalk is alone in the top-left corner on airports and wiki-RfA; on directed motifs, directed ReFeX is more accurate and 40 times slower. `speed_figure.py`.

Table 4: Wikipedia talk networks (RONE): precision@10 for bots and admins, mean over the 6 networks, and over the 3 smallest (446–2,101 nodes), the only ones struc2vec runs on; only two of these have admins (8 and 16), so the "Admins (3)" column rests on 24 queries. `roles.csv`.

	Bots (6)	Admins (6)	Bots (3)	Admins (3)
RoleWalk	**0.129**	**0.324**	**0.179**	0.356
RoleWalk + curv.	0.121	0.322	0.162	0.356
struc2vec	–	–	0.146	**0.391**
GraphWave	0.115	0.298	0.163	0.341
RESD	0.114	0.254	0.158	0.334
ReFeX	0.101	0.299	0.161	0.375
Role2Vec	0.119	0.016	0.158	0.041
RWSE	0.118	0.297	0.178	0.309
Degree (baseline)	*0.075*	*0.296*	*0.121*	*0.341*

RoleWalk leads on bots and on admins over all networks. Where struc2vec runs, its lead on admins comes from br-wiki (8 admins; per-query paired Wilcoxon $p = 0.06$ against RoleWalk, $p = 0.04$ for ReFeX, uncorrected); exact walks, a mass budget, other walk lengths, wider frequency ranges, local statistics or whitened distances do not change it. Admins differ from other hubs by low clustering and low neighbour degree: they write to many newcomers who do not write to each other. Networks are distributed undirected.

Table 5: Directed roles on synthetic graphs (directed motifs hung from a directed cycle; several roles coincide once directions are dropped). 3 seeds; noise = share of edges rewired. Accuracy: logistic regression; ARI: k -means; mAP: retrieval. ReFeX, directed: in-, out- and mutual degrees and recursive features over out- and in-neighbours (450 one-hot features). Top-k + layers reads the out- and in-walks, not the mixed probes: compare it with RoleWalk without them (note). `directed.csv`.

	Acc. 0%	Acc. 5%	Acc. 10%	ARI 5%	mAP 5%
RoleWalk + curv.	**1.000**	0.730	0.593	0.396	**0.629**
RoleWalk	**1.000**	0.728	0.588	0.373	0.615
ReFeX, directed	**1.000**	**0.756**	**0.638**	0.355	0.607
Digraphwave	**1.000**	0.717	0.575	0.292	0.539
Local statistics (in / out)	0.863	0.675	0.584	**0.409**	0.609
Top-k + layers	0.908	0.671	0.516	0.348	0.568
RWSE, out + in	0.357	0.336	0.336	0.048	0.179
Role2Vec	0.732	0.440	0.373	0.153	0.296
ReFeX	0.532	0.429	0.431	0.342	0.356
GraphWave	0.586	0.388	0.383	0.317	0.416
RESD	0.547	0.420	0.423	0.253	0.348
struc2vec	0.510	0.366	0.277	0.141	0.260
In/out degree (baseline)	*0.516*	*0.466*	*0.405*	*0.310*	*0.349*

Ablations. Without the mixed probes: RoleWalk + curv. 0.989 / 0.717 / 0.547 / 0.381 / 0.625, RoleWalk 0.908 / 0.682 / 0.510 / 0.366 / 0.580 (never better on any seed and noise level for RoleWalk; within noise for RoleWalk + curv.). On the symmetrised graph: RoleWalk 0.767 / 0.560 / 0.420 / 0.351 / 0.474. Classifier: over 10 splits per graph, with gradient boosting instead of logistic regression, RoleWalk + curv. reaches 1.000 / 0.744 / 0.586 and directed ReFeX 1.000 / 0.739 / 0.578 (logistic: 1.000 / 0.741 / 0.585 against 1.000 / 0.753 / 0.610), so the two are comparable on these motifs; no other method changes rank (`directed_gb*.csv`).

Table 6: Real directed roles: Wikipedia requests for adminship [22] (11,377 users, 188k votes voter → candidate; signs dropped, repeated votes merged, unweighted). A: elected among the 3,494 candidates. B: voter / failed candidate / admin. `roles_rfa*.csv`.

	A: elected	B: 3 classes	Time
RoleWalk + curv.	**0.827**	**0.944**	7 s
RoleWalk	0.814	0.940	7 s
RESD	0.798	0.843	68 s
Role2Vec	0.798	0.857	82 s
ReFeX	0.794	0.847	1 s
ReFeX, directed	0.778	0.930	3 s
Digraphwave	0.807	0.939	360 s
Top-k + layers	0.804	0.936	5 s
Local statistics (in / out)	0.814	0.941	3 s
RWSE, out + in	0.710	0.836	84 s
In/out degree (baseline)	*0.755*	*0.922*	*0 s*

struc2vec and GraphWave do not run at this size. Ablations. Without the mixed probes: RoleWalk + curv. 0.823 / 0.943 (5 s), RoleWalk 0.808 / 0.938 (4 s); on the symmetrised graph: 0.814 / 0.900 and 0.791 / 0.863. RWSE on the out-walk only: 0.690 / 0.826. Paired over the 50 splits (Nadeau–Bengio, `paired_rfa.csv`), RoleWalk + curv. is significantly better than every other method on A (local statistics $+0.012$ $[0.001, 0.022]$, Digraphwave $+0.019$ $[0.008, 0.029]$) and on B, except local statistics in accuracy ($+0.003$ $[-0.000, 0.006]$; macro-F1 $+0.007$ $[0.000, 0.014]$).

Table 7: Trophic roles in 20 real food webs [23] (1,086 compartments, 20–125 per web; edge prey → predator, weighted by carbon flow; boundary nodes and self-loops removed). Labels: 8 trophic guilds (producer, detritus, microbial, herbivore, detritivore, omnivore, invertivore, top predator), proxy labels given by LLMs from the compartment names and the ecosystem alone, without the graph (`foodweb_labels.py`): GPT-6 (low effort), GPT-6 (high) and Claude Opus 5.5 (high) agree on 88.6% of the compartments (pairwise Cohen's κ 0.88–0.96); disagreements are ecologically ambiguous (oysters, copepods). Leave one ecosystem out (12 ecosystems; seasons and sub-webs of one ecosystem stay together). Balanced accuracy, mean over ecosystems. `foodwebs*.csv`.

	Flows	Unweighted, one labelling	Majority of 3	Unanimous (88.6%)
RoleWalk	0.428	0.524	0.517	**0.569**
RoleWalk, no mixed probes	0.420	0.459	0.465	0.456
RoleWalk + curv.	0.413	0.492	0.515	0.543
Top-k + layers	0.377	0.489	0.487	0.514
Digraphwave	**0.429**	**0.559**	**0.558**	0.547
Local statistics (in / out)	–	0.527	0.540	0.565
ReFeX (frozen)	–	0.232	0.234	0.251
RWSE, out + in (flows)	0.214	0.214	0.214	0.210
In/out degree and flow (baseline)	*0.436*	*0.436*	*0.447*	*0.455*

Carbon flows span 19 orders of magnitude; read as transition weights they send walks along the few largest flows (to detritus), and every walk-based method evaluated both ways is better on the unweighted web, where the guild is who eats whom. On unweighted webs, RoleWalk, the local statistics and Digraphwave are not distinguishable: their ranking changes with the labelling and no paired difference is significant (RoleWalk minus local statistics -0.003 $[-0.08, 0.08]$, -0.023 $[-0.10, 0.06]$, $+0.004$ $[-0.05, 0.06]$; minus Digraphwave -0.035, -0.041, $+0.022$, all intervals including 0). Robust to the labelling: the mixed probes add $+0.065$ $[0.006, 0.124]$, $+0.052$ $[-0.015, 0.120]$, $+0.114$ $[0.046, 0.181]$ (sharing prey or predators, i.e. competition, is a trophic role), and RoleWalk is far above ReFeX and RWSE. Top-k + layers reads the out- and in-walks without the mixed probes. RoleWalk on the symmetrised flow web: 0.361.

Table 8: Expressiveness (tol = 0): pairs told apart (CSL: number of the 10 isomorphism classes told apart). Every set is 1-WL-indistinguishable. Two graphs are told apart when their multisets of node embeddings differ beyond float32 noise (sorted rows compared at a relative tolerance 10^{-4}). No false difference on random relabellings. `expressiveness.csv`.

	Atlas ≤ 7 (20)	Regular (3,181)	CSL (10 cl.)	Decalin / bicyc.	$2C_3$ / C_6	SRG pair*
RoleWalk (3 steps)	**20**	**3,181**	6	yes	yes	no
RoleWalk (5 steps)	**20**	**3,181**	**10**	yes	yes	no
RWSE (16 steps)	**20**	3,180	**10**	yes	yes	no
Degree (baseline)	*0*	*0*	*1*	*no*	*no*	*no*

* Shrikhande / 4×4 rook: strongly regular with equal parameters, hence not separated by Proposition 7. Without layers, RoleWalk (3 steps) separates 16 / 3,180 / 6.

Table 9: BREC [17] (400 pairs; tol = 0).

	Basic (60)	Regular (100)	Extension (100)	CFI (100)	4-vertex (20)	Dist.-reg. (20)	Total
RoleWalk (3 steps)	60	50	97	0	0	0	207
RoleWalk (5 steps)	60	50	100	3	0	0	213
RWSE (16 steps)	60	50	100	4	0	0	**214**

Distance-regular: 0 by Proposition 7. The 50 regular pairs missed are exactly the 50 strongly regular pairs, all with equal parameters (checked). Without layers, RoleWalk (3 steps) separates 203 pairs.

Table 10: Centrality transfer to larger graphs: a gradient-boosting regressor from features to centrality is fitted on a 2,000-node graph and evaluated (Spearman) on a 8,000-node graph of the same family at the same mean degree (other seed); 6 families (Barabási-Albert, powerlaw-cluster, Watts-Strogatz, SBM, random geometric, motifs) × 10 independent pairs of graphs. Path-based centralities and PageRank (top four rows) are where walks matter. † The clustering coefficient is one of the features of the local statistics and of ReFeX (a base feature that ReFeX aggregates recursively), which explains their lead on this row. 20 of the 420 test graph–target cells have a constant target (k-core: all 10 Barabási-Albert graphs, 9 powerlaw-cluster, 1 Watts-Strogatz) and are left out. Table means average the 400 valid cells; paired differences average the targets of each pair of graphs, then the pairs. ReFeX's feature selection, scaling and bins are fitted on the training graph and frozen. Walk length: RoleWalk, its variants and top-k + layers use 4 steps here (94 features each; RoleWalk exact, tol = 0; top-k pruned at 10^{-5}), not the default 3. `centrality_transfer_x4_10seeds.csv`, `transfer_paired.csv`, `transfer_topk4/`.

	RoleWalk	+ curv.	Top-k + layers	Local stats	ReFeX (frozen)	RW heat CF	RWSE	GraphWave, no $1/n$	*Degree*
Betweenness	0.864	**0.866**	0.859	0.852	0.845	0.849	0.412	0.659	*0.623*
Closeness	**0.797**	0.793	**0.797**	0.716	0.725	0.738	0.464	0.529	*0.407*
Harmonic	**0.861**	0.860	0.856	0.778	0.788	0.806	0.517	0.597	*0.460*
PageRank	**0.989**	**0.989**	**0.989**	0.984	0.973	0.986	0.741	0.611	*0.929*
Clustering	0.805	0.802	0.794	**0.844**†	0.813†	0.785	0.723	0.347	*0.201*
k-core	0.651	0.652	**0.653**	0.650	0.631	0.647	0.494	0.594	*0.614*
Eigenvector	0.673	0.671	0.680	0.661	**0.700**	0.586	0.358	0.415	*0.463*
Mean	**0.813**	0.812	0.812	0.790	0.790	0.777	0.532	0.533	*0.524*

Local stats: degree, neighbour-degree moments, triangles, egonet boundary, clustering, lazy-walk returns at steps 2 and 3. RW heat CF: random-walk heat-kernel characteristic function. Component attribution, same walks (`transfer_attr/`): layers alone (14 features) 0.807, characteristic function alone 0.785, lazy return probabilities (RWSE on H, 4 steps) 0.753; RoleWalk minus layers alone +0.007 $[0.004, 0.009]$ (78% of pairs), minus CF alone +0.029 $[0.024, 0.035]$. The law of the distance carries most of the transfer; the CF adds a small, consistent gain over the layers, and the top-k probabilities of the same four steps do as well as the CF (0.812). An earlier version of this table compared four-step RoleWalk with three-step top-k (0.801); that gap came from the extra step. Without the clustering target, means: RoleWalk 0.815, top-k + layers 0.815, frozen ReFeX 0.785, local stats 0.781; paired, RoleWalk minus local stats +0.034 $[0.026, 0.041]$. Under a change of generator parameters instead of size: Section 6.4. Paired over the 60 pairs of graphs (RoleWalk minus other, mean over targets, 95% t-interval, share of pairs won): top-k + layers +0.002 $[0.000, 0.004]$, 67%; local stats +0.022 $[0.016, 0.029]$, 80%; frozen ReFeX +0.024 $[0.018, 0.029]$, 82%; RW heat CF +0.038 $[0.030, 0.046]$, 98%. On closeness and harmonic, RoleWalk wins every pair against local stats and ReFeX (+0.07 to +0.08) and ties top-k + layers (closeness +0.001 $[-0.003, 0.005]$, harmonic +0.005 $[0.003, 0.007]$, betweenness +0.004 $[-0.001, 0.010]$); local stats win clustering (−0.04) and ReFeX eigenvector (−0.03). Per family (mean over targets; RoleWalk / + curv. / top-k / local stats / ReFeX / heat CF): Barabási-Albert 0.881 / 0.882 / 0.879 / 0.866 / 0.845 / 0.792; powerlaw-cluster 0.925 / 0.925 / 0.921 / 0.928 / 0.923 / 0.863; random geometric 0.577 / 0.571 / 0.572 / 0.532 / 0.531 / 0.557; SBM 0.829 / 0.831 / 0.826 / 0.813 / 0.796 / 0.813; motifs 0.881 / 0.876 / 0.883 / 0.880 / 0.883 / 0.853; Watts-Strogatz 0.812 / 0.813 / 0.812 / 0.750 / 0.785 / 0.799. With 2 seeds, test graphs of the training size give the same order (means: RoleWalk 0.851, three-step top-k + layers 0.835, ReFeX 0.822, RW heat CF 0.819, local stats 0.817, RWSE 0.726, GraphWave 0.633; `centrality_transfer.csv`): descriptively (2 against 10 seeds), from the same size to 4 × the size, RoleWalk's mean drops by 0.04, RWSE's by 0.19 and GraphWave's (no $1/n$) by 0.10. GraphWave's scale is chosen from each graph's spectrum (Proposition 11); RWSE's coordinates are local like RoleWalk's, so its drop is not a matter of coordinates, and we have not diagnosed it. With 2 seeds, appending the local statistics to RoleWalk gives 0.875 / 0.780 / 0.863 / 0.990 / 0.838 / 0.657 / 0.654, mean 0.808. Digraphwave does not transfer (2 seeds, mean 0.08; `centrality_transfer_x4_dgw.csv`): its coordinates are near 1 with spreads that shrink with the graph's size, so on the larger graph they fall outside the bins learned on the smaller one and the regressor predicts a constant on 25 of the 80 scored pairs (counted as 0). With an unfrozen ReFeX (only where feature counts happen to match), k-core reached 0.935 at the same size: an artefact of the subset.

Table 11: Role transfer between airport networks: classifier fitted on one network, tested on another (6 ordered pairs, accuracy). Labels are activity quartiles, close to degree: this transfer is won by degree-level statistics, and the centrality transfer of Table 10 is the relevant test of what walks transfer. Removing GraphWave's $1/n$ lifts it from chance to 0.455 (Proposition 11); fixed scales and a heat kernel instead of walk powers give 0.425. `role_transfer.csv`, `role_transfer_gw.csv`.

RoleWalk	+ curv.	Top-k + layers	Local statistics	ReFeX (frozen)
0.488	0.472	0.475	**0.565**	0.438
RWSE	GraphWave	GraphWave, no $1/n$	RW heat CF	*Degree (baseline)*
0.375	0.249 (chance)	0.455	0.425	*0.542*

Table 12: Structural encodings for GNNs on ZINC (test MAE at the best validation epoch). Top: GINE, 4 layers, ≈230k parameters, 300 epochs, 3 seeds. Bottom: GPS [12] protocol, 10 layers, ≈470k parameters, 500 epochs, 5 seeds. `pe_zinc_v13.csv`, `pe_zinc_gps*.csv`.

GINE	Test MAE	
none	0.202 ± 0.010	
RWSE	0.107 ± 0.004	
RoleWalk	0.108 ± 0.008	
RoleWalk + RWSE	0.104 ± 0.006	
RoleWalk + curv. + RWSE	**0.095 ± 0.002**	
GPS	Test MAE	Val. MAE
RWSE (published: 0.070 ± 0.004)	0.077 ± 0.004	0.100 ± 0.005
RoleWalk + curv.	0.078 ± 0.006	0.093 ± 0.003
RoleWalk + curv. + RWSE	**0.072 ± 0.001**	**0.092 ± 0.006**

RoleWalk + curv. + RWSE is below RWSE on each of the 5 seeds (paired difference 0.0045, standard deviation 0.0037 over seeds, 95% interval $[-0.0001, 0.0091]$; two-sided sign test $p = 0.06$). RoleWalk + curv. alone is unstable across seeds (0.073 to 0.089).

Table 13: Embedding time against size with the fused kernel, defaults (tol 10^{-4} without budget, 3 steps, layers), including the conversion of the networkx graph; one run after compiling the kernel, Intel Core i9-9900X (3.5 GHz). `scalability.csv`.

Nodes	10 k	30 k	100 k	300 k	1000 k
Barabási-Albert ($m = 3$), 1 core	0.47 s	1.84 s	8.94 s	43.05 s	192.93 s
+ curvature	0.56 s	1.94 s	11.22 s	52.36 s	246.86 s
8 threads	0.22 s	0.75 s	6.34 s	13.69 s	72.7 s
Watts-Strogatz ($k = 6$), 1 core	0.12 s	0.45 s	1.78 s	5.11 s	17.65 s
+ curvature	0.16 s	0.48 s	1.77 s	5.39 s	19.16 s
8 threads	0.06 s	0.31 s	0.81 s	2.44 s	9.78 s

Peak memory 2.5 GB at a million nodes, of which 1.6 GB hold the networkx graph. Baselines on the same Barabási-Albert graphs, one core (`scalability_baselines.csv`): ReFeX 0.1 s / 1.8 s / 31 s at 10k / 100k / 1M nodes, faster than RoleWalk; 16-step RWSE 68 s at 10k and over 16 GB at 100k; RESD 62 s / 631 s at 10k / 100k and over an hour at 1M; Role2Vec 56 s at 10k and over 10 GB at 100k; the heat-kernel CF 196 s and Digraphwave 184 s at 10k, both over an hour at 100k. RoleWalk is the only embedding evaluated, other than ReFeX, to reach a million nodes. At 1,190 nodes (USA airports): struc2vec 564 s, GraphWave 18 s, Role2Vec 10 s, RESD 8 s; struc2vec and GraphWave do not run on wiki-RfA (11k nodes).

6 Analysis

6.1 Summaries of the walk distribution

Table 14: Other summaries in place of the characteristic function, without layers or curvature (an earlier ablation, not dimension-matched: top-10 probabilities give 10 coordinates per step against 20 for the characteristic function; the dimension-matched top-20 + layers is in Table 2, Table 5 and Table 10). Airports: mean accuracy; Transfer: airport transfer; Motifs: synthetic accuracy.

	Airports	Transfer	Motifs (varied / houses)	Time 20k
Characteristic function	0.656	0.445	**0.744** / 0.733	6.5 s
Laplace transform	**0.657**	0.345	0.614 / 0.717	18 s
Dyadic histogram (Haar)	0.631	0.353	0.614 / 0.725	15 s
Power sums (Rényi)	0.653	0.420	0.647 / 0.692	9.7 s
Top-10 probabilities	0.652	**0.512**	0.679 / **0.758**	3.5 s

6.2 Geometry: moments rather than integrated characteristic functions

Table 15: Ways to add local geometry (earlier ablation runs; directed accuracy without the mixed probes, against 0.728 with them in Table 5). Here, degree marks and counting bounds hurt transfer.

	Airports	Transfer	Motifs (varied / houses)	Directed acc.
No geometry	0.656	0.445	0.744 / 0.733	0.678
+ onion layers (moments)	0.661	<u>0.488</u>	0.744 / <u>0.758</u>	0.682
+ layers, one CF per layer	0.654	0.424	<u>0.751</u> / 0.742	0.695
+ degree marks (joint CF)	0.65–0.67	0.34–0.36	–	–
+ curvature, counting bound	<u>0.667</u>	0.388	**0.754** / **0.767**	<u>0.721</u>
+ curvature, probabilistic (moments)	**0.668**	0.470	0.725 / 0.742	0.712
+ curvature, exact Ollivier-Ricci	0.662	**0.494**	0.746 / 0.750	**0.743**
+ curvature, integrated CF	0.663	0.400	<u>0.751</u> / 0.750	0.697

The probabilistic bound is kept over exact Ollivier-Ricci for cost (no optimal transport) at a small loss.

Marks on the steps of a path rather than on its end were also tried: the parity of steps inside a layer (closing odd cycles), obtained from one extra signed walk $\tilde{D}^{-1}(A - I)$ and the layer of arrival, since on undirected graphs $\#\text{intra} \equiv \#\text{moves} - d(u, j) \pmod 2$; its exact count; and the order of moves out and in, as products in the quaternion group Q_8. On the 6,781 nodes of connected graphs with at most 7 nodes, none separates a pair of nodes that X does not, and none improves airport accuracy (`intra_layer_study.py`, `quaternion_study.py`). With labels g (out) and g^{-1} (in) in an abelian group and trivial labels otherwise, the product along a path only depends on its end layer, which gives back the layers; short cycles already change the multiset of probabilities.

A purer object was also tried: one characteristic function of each node's whole trajectory $\left(1[j = u], p^1_{uj}, p^2_{uj}, p^3_{uj}\right)$, sampled along marginal and mixed-time projections (walks stopped at a random time), in place of per-step functions and layers. It sees whether large probabilities at different steps sit on the same node, which per-step multisets forget, and makes every coordinate Lipschitz. At 48 to 160 features it separates the same small-graph nodes as RoleWalk and matches it on airports (0.664–0.666 against 0.662), but transfers worse (0.454–0.471 against 0.486) and loses on directed motifs (ARI 0.295–0.314 against 0.366, all without mixed probes) (`trajectory.csv`): hard layers are a better use of a few dimensions.

6.3 An observation: betweenness from one small graph

A regressor fitted once on a single 2,000-node graph (exact betweenness, about a second of training) predicts the betweenness ranking of new graphs of the same family up to 25 times larger, from RoleWalk alone, without computing any shortest path on them (Table 16).

Table 16: Spearman correlation with exact betweenness, and time on one core (embedding and prediction), for a regressor fitted on one 2,000-node graph and applied to new graphs of the same family; mean of 3 seeds (1 at 50k). `betweenness.csv`.

Test graph	5,000 nodes	20,000 nodes	50,000 nodes
Barabási-Albert	0.953 (0.7 s)	0.937 (3.7 s)	0.925 (11.7 s)
Powerlaw-cluster	0.926 (0.7 s)	0.907 (3.5 s)	0.896 (11.4 s)
Watts-Strogatz	0.960 (0.2 s)	0.963 (1.1 s)	0.962 (2.8 s)

The ranking degrades slowly with size on hub-dominated families and not at all on Watts-Strogatz. This is not a substitute for dedicated algorithms: exact Brandes (NetworKit) takes 163 s at 20,000 nodes on one core, but source sampling reaches similar correlations at similar cost, and adding local statistics to RoleWalk brings it to the same level (0.936–0.964 at 50k). What it shows is that a few walk statistics, read the same way in every graph, carry most of a global, path-based quantity across sizes.

6.4 Transfer under parameter shifts

Table 10 changes the size of the graph, not its generator. Here the regressor is fitted on one 2,000-node graph and tested on a 2,000-node graph whose generator has one parameter changed, fixed in advance:

Barabási-Albert $m = 3 \to 5$ (mean degree), powerlaw-cluster triangle probability $0.3 \to 0.6$ (clustering), Watts-Strogatz rewiring $0.1 \to 0.3$, SBM within-block probabilities doubled, random geometric radius $0.04 \to 0.06$ (density); 10 independent pairs each, same features and protocol.

Table 17: Centrality transfer under a parameter shift (Spearman; per target, mean over the 50 pairs; overall, mean of the 330 valid graph–target cells, 20 cells with a constant k-core left out: all 10 Barabási-Albert, 9 powerlaw-cluster, 1 Watts-Strogatz; paired intervals in the text average targets within each pair, then pairs). RoleWalk and top-k + layers: 4 steps, 94 features each; top-k from a separate run, against which RoleWalk recomputed with the current library scores 0.635 (floating-point differences between library versions). `centrality_transfer_shift.csv`, `shift_paired.csv`, `transfer_shift_topk4/`, `shift_topk4_paired.csv`.

	RoleWalk	Top-k + layers	Local stats	ReFeX (frozen)	RW heat CF	RWSE	GraphWave, no $1/n$	*Degree*
Betweenness	0.614	0.587	**0.742**	0.701	0.614	0.373	0.619	*0.644*
Closeness	**0.720**	0.686	0.618	0.597	0.599	0.349	0.490	*0.420*
Harmonic	**0.750**	0.704	0.657	0.624	0.663	0.367	0.567	*0.496*
PageRank	0.893	**0.929**	0.923	0.831	0.677	0.625	0.878	*0.886*
Clustering	0.639	0.630	**0.959**[†]	0.759[†]	0.539	0.640	0.040	*−0.000*
k-core	−0.017	−0.112	0.134	0.018	−0.065	0.012	**0.235**	*0.485*
Eigenvector	**0.612**	0.558	0.605	0.554	0.533	0.360	0.513	*0.514*
Mean	0.639	0.610	**0.695**	0.618	0.543	0.412	0.492	*0.493*

[†] Clustering is one of the features of these methods, which explains their lead on this row. Without the clustering target, means: RoleWalk 0.639, local stats 0.647, frozen ReFeX 0.593, top-k + layers 0.607; paired, RoleWalk minus local stats -0.008 $[-0.033, 0.017]$, minus frozen ReFeX $+0.049$ $[0.020, 0.078]$.

RoleWalk has the highest mean among the embedding baselines. Paired over the 50 pairs, its lead is established over the dimension-matched top-k + layers ($+0.024$ $[0.003, 0.044]$, against a tie under a change of size), the heat-kernel CF, RWSE and GraphWave, but not over frozen ReFeX ($+0.023$ $[-0.007, 0.054]$). It has the best means on closeness and harmonic; against local statistics the gaps are established ($+0.10$ $[0.05, 0.16]$ and $+0.09$ $[0.04, 0.15]$), against top-k + layers on harmonic only ($+0.046$ $[0.008, 0.084]$; closeness $+0.029$ $[-0.003, 0.061]$). The local statistics are better on average (-0.055 $[-0.080, -0.031]$), mostly on clustering (-0.32), which is one of their features (without it: -0.008 $[-0.033, 0.017]$), and on two global quantities that they predict better from local features, k-core (-0.15; the degree alone is best) and betweenness (-0.13). Per family, RoleWalk's mean is 0.911 / 0.899 / 0.807 on Barabási-Albert, powerlaw-cluster and Watts-Strogatz, and every method falls to 0.1–0.5 on SBM and random geometric, where changing density or community strength changes what the centralities measure. Across the tested within-family parameter shifts, walk features transfer closeness and harmonic centrality best; a feature that is the target (clustering), or degree-driven quantities (k-core, betweenness), transfer better from local statistics.

6.5 Pruning

On the six real networks, pruning at the default tol 10^{-4} changes accuracy by at most 0.005 (0.002 with a budget), and by at most 0.012 at tol 10^{-3} (`pruning.csv`). Without a budget, the characteristic-function block keeps a median relative error below 10^{-4} at tol 10^{-4}, but layers move by up to 16% (73% at tol 10^{-3}). The error concentrates on hubs: for the centre of a star with 20,000 leaves and the largest hub of a 100k-node Barabási-Albert graph, the embedding is off by 49% and 27%. A budget δ caps the worst absolute ℓ_1 error of the unscaled layers at each step at δ (observed 0.009, 0.047, 0.096 for $\delta = 0.01, 0.05, 0.10$) and makes both hubs exact; without curvature, the observed feature error is at most 0.51 of the certificate of Proposition 13 everywhere. On the real networks the budget costs nothing measurable (3.3 s against 3.1 s in total); on the hub-heavy Barabási-Albert graph it doubles the time, as hub rows are kept whole. Table 18 measures the trade-off on hub-heavy and directed graphs. With the fused kernel, exact walks (tol $= 0$) take seconds at 50k nodes; threshold pruning buys up to 8× in time at a small median but large worst-case error, and a budget buys certified, near-exact features at about the exact cost. The library returns each node's missing masses and certificate, so a user can check the worst case of a pruned run instead of trusting the median. At a million nodes (Barabási-Albert, one core, tol 10^{-4}; `certified_scale.csv`), threshold pruning alone takes 173 s but leaves 69% of the rows missing more than 1% of their mass (some all of it; largest certificate 366);

a budget of 1% takes 227 s (1.3×) and 1.7 GB, with every certificate below 3.9. On Watts-Strogatz graphs no mass is lost (17 s at a million nodes).

Table 18: Certified pruning as a setting (curvature-free): one core, peak memory of the process (graph included), relative error of each node's embedding against the exact one (median and worst node, %), and the largest per-node certificate (`RoleWalk.certificate_`, absolute; $\|X_u\|$ is a few hundred). Directed SF: directed scale-free graph (Bollobás et al.), mixed probes included. `pruning_frontier.csv`.

	tol, budget	Time	GB	Median %	Worst %	Cert. max
BA 50k	exact	3.8 s	0.29	0.0	0.0	–
BA 50k	10^{-4}	3.0 s	0.30	1.3	22.4	188
BA 50k	10^{-3}	2.3 s	0.32	11.0	48.1	367
BA 50k	10^{-4}, $\delta = 0.01$	3.6 s	0.31	0.0	0.7	4
BA 50k	10^{-3}, $\delta = 0.05$	3.5 s	0.32	0.0	3.9	20
Directed SF 50k	exact	38.2 s	0.71	0.0	0.0	–
Directed SF 50k	10^{-4}	7.8 s	0.61	0.6	44.9	414
Directed SF 50k	10^{-3}	4.9 s	0.53	4.4	46.7	487
Directed SF 50k	10^{-4}, $\delta = 0.01$	43.1 s	0.76	0.0	0.6	3
Directed SF 50k	10^{-3}, $\delta = 0.05$	42.3 s	0.76	0.0	3.4	26
wiki-RfA	exact	4.1 s	0.61	0.0	0.0	–
wiki-RfA	10^{-4}	3.6 s	0.58	1.6	6.7	77
wiki-RfA	10^{-3}	2.2 s	0.54	11.2	34.9	424
wiki-RfA	10^{-4}, $\delta = 0.01$	4.3 s	0.59	0.0	0.5	5
wiki-RfA	10^{-3}, $\delta = 0.05$	4.3 s	0.59	0.0	2.5	20

Observed error over certificate: at most 0.64 on every node of every run (0.45 in all but one). With a budget, rows that would exceed it keep all their entries, so a budget of 1% is nearly exact at the cost of the exact time: on the directed graph, the budget is even slower than exact computation (not profiled; mixed products are formed from H_+ and H_- and pruned afterwards, so retained pure-walk rows do not feed them). Threshold pruning alone is 1.2–8× faster, with a small median error but up to 48% on the worst node.

6.6 Kernel scaling

In raw embeddings $\theta = 100$ carries 53% of the variance; unscaled layers carry 0.02% and leave k-means roles unchanged. A grid of 48 configurations (frequency weighting, range, steps, layers), scored on synthetic and real networks with equal weight (`kernel_grid.csv`): scaling the layers by $\|\Theta\|$ is what matters (score 0.274 $\rightarrow$ 0.323), the frequency weighting barely does, and the defaults win whichever dataset is left out. Spacing matters less than range (`theta_scheme.csv`): on $[1, 100]$, log-spaced frequencies score 0.323 against 0.316 for uniform ones, with the same airport accuracy (0.661 / 0.662) and transfer (0.488 / 0.485). A narrower range helps real networks and transfer but loses on synthetic motifs, whatever the spacing: $[1, 30]$ gives transfer 0.515 (log or uniform) and score 0.312–0.315; log $[1, 50]$ gives 0.506 and 0.326, within noise of the default. Log spacing is kept as a heuristic: probabilities spread over several decades, which a log grid samples evenly, while a uniform grid puts 9 of its 10 frequencies above 10. Feature-wise standardisation, as in our classifiers, cancels any constant factor on a block: the scaling of the layers matters for raw distances (k-means, Ward, nearest neighbours), not for the classification tables.

6.7 Film is a node-type task

Table 19: Film (RONE). Class 3 is an independent set in a separate block of node ids.

	Class 3 vs rest	Within classes 0–2
Role2Vec	**0.946**	0.637
RoleWalk + curv.	0.740	<u>0.634</u>
ReFeX	<u>0.745</u>	0.627
Degree (baseline)	*0.536*	*0.623*
Majority class	0.593	0.623

No method predicts classes 0–2 clearly above the majority class; edge homophily is 0.28, below the 0.32 expected from random labels with the same (degree-weighted) class proportions. Role2Vec's lead on class 3

is consistent with walk co-occurrence, i.e. position, a hypothesis we have not tested (a node-ID permutation check would only rule out identifier artefacts); we read Film as a node-type task rather than a role benchmark.

7 Limitations

- Walk statistics cannot separate distance-regular graphs with equal arrays (Proposition 7), most CFI pairs (3 of 100 with 5 steps), or graphs whose balls coincide (locality); without curvature, some nodes separated by 1-WL are confused (Proposition 9).
- Without a mass budget, the pruning certificate is only a posteriori and large on hubs (Section 6.5); the budget costs time on hub-heavy graphs, and has no support cap.
- ReFeX is faster (5× at a million nodes) and wins on noisy synthetic directed motifs in supervised accuracy (with logistic regression; comparable with gradient boosting); a fixed-scale heat-kernel CF matches RoleWalk on two of three airport networks.
- Middle-tier roles on real networks often form a continuum; per-node stability should be reported next to roles.
- Degree-like labels (airports) favour trivial baselines.
- Under a change of generator parameters (Section 6.4), hand-made local statistics transfer better than RoleWalk on average (clustering, k-core, betweenness); RoleWalk still leads every embedding. Under a change of size, a dimension-matched top-k summary of the same walks ties RoleWalk (Table 10).
- Two real labelled directed settings (wiki-RfA, food webs); on food webs (proxy labels from three LLMs, 88.6% unanimous) RoleWalk, local statistics and Digraphwave are not distinguishable, and raw flow weights hurt every walk-based method evaluated both ways.
- GPS on ZINC: 5 seeds; our RWSE reproduction (0.077) is weaker than the published 0.070; RoleWalk + curvature + RWSE (0.072) is better on every seed, but the paired 95% interval just includes 0 (sign test $p = 0.06$).

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